

A Search for Extended Circumgalactic Dust

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1 Motivation

- Motivate the research based on literature results: obviously Menard, but also Peek, and the theory papers. Do not include figures, except maybe in an appendix
- DO NOT CONGRATULATE
- Should probably explain how A_v , reddening and R_v are related to one another
 - Linearly related! But dust-model dependent
 - We chose $R_v = 3.1$ as a default, but others may be better...
 - * consider moving this discussion to methods?

2 Methods

2.1 A Maximum Likelihood Estimator for Dust Extinction

- Develop our MLE formalism; reproduce equations
 - don't forget to include discussion of extinction vector (FM07 or whatever)
- May also want to include the ON and optimal vector generation, at least for a paragraph –YES
 - Think of this like a PCA; as long as the sum is non-negative, we're in good shape.
 - Dips positive or negative mean that the “principal component” under or overfits the actual trend at that distance
 - * little harder to connect to physical meaning for now – problem for theorists?
 - For development of “optimal” vector, mb show chi-squared plots, just to explain where we got this?

* I think I baled those, but fairly easy to reproduce

- Discuss how and why we split A_V estimate into redshift bins
 - correlating *deviations from mean color space of background galaxies*
 - **HEY: did we deredden our magnitudes for galactic reddening? Are we hoping that we just "average" over dust? Because that's a bad idea**
- yooo need to justify $R_V = 3.1$ and consider using other values of R_V

2.2 Correlation Function Algorithm

- Treecorr algorithm, briefly: it's not my algorithm but it is what we use. Correlating not gal-shear but gal-extinction
- Mention spatial correlation as well
- Mention null tests: flexibility of algorithm allows us to easily correlate any two samples, so ideal for null tests like star-extinction

3 Datasets

- Discuss samples that we used:
 - redMaGiC as background (discuss algorithm, add some plots of color variance with MLE or color-color plots)
 - * area covers: $290^\circ \lesssim \alpha \lesssim 110^\circ$
 - GALEX and IIFSCz (maybe)
- Also SVM for DES foreground sample
 - give some characteristics
 - perhaps move discussion of algorithm to ??
 - * ref needed
- Might be good to have some consistency plots showing overlap of two samples
- Used Healpy to ensure that we are not correlating across empty space, which would have the effect of diluting the observed signal

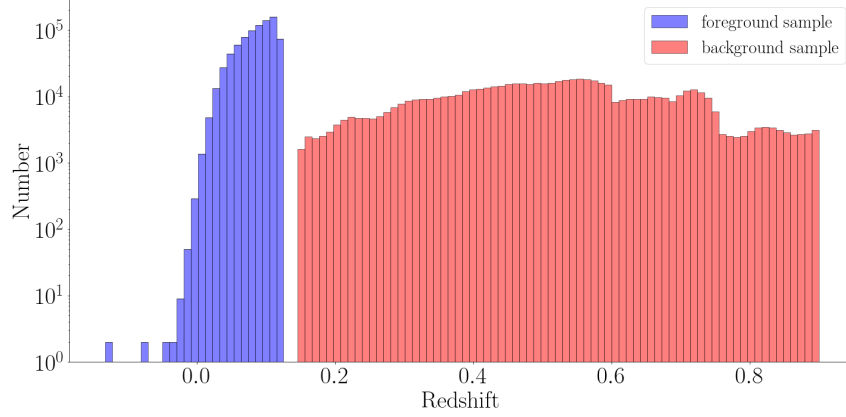


Figure 1: Redshift distribution of foreground and background samples

4 Results

- Plot up all the results I do have... with dust vector, "optimal" vector for all samples
- MB show the "dust mass plot" but maybe also skip because it's not really a dust mass
- Star-bkg correlation is a good null test to show

5 Conclusions

- Start discussing consequences: here, compare to literature
- It might be cool to compare to MgII absorption plot, since that also shows evidence for extinction
 - Except that maybe that's controversial for actual MgII folks? Apparently they expect this to be anti-correlated
- For the literature figures, consider placing in appendix?